

EvoBin Deployment Guide

Production Deployment Protocol: MongoDB Atlas + Render + Vercel

Architecture & Prerequisites

- **Frontend:** React + Vite + TypeScript (Hosted on **Vercel**)
- **Backend:** FastAPI + Uvicorn + OpenCV Headless (Hosted on **Render**)
- **Database:** MongoDB Atlas Cloud Cluster
- **Version Control:** Git & GitHub Repository

Codebase Fixes Applied

File Path	Modification	Purpose
frontend/src/services/api.ts	Dynamic API Base URL	Uses import.meta.env.VITE_API_BASE_URL
frontend/vercel.json	SPA Rewrite Rule	Prevents 404 errors on browser page refresh
backend/requirements.txt	Headless OpenCV	Fixes missing Linux libGL.so.1 dependency on Render
backend/Procfile	Render Web Process	Uvicorn startup: web: uvicorn app.main:app
backend/app/config.py	Environment Defaults	Provides safe defaults for DB_NAME

Step 1: MongoDB Atlas Setup (Database)

1. **Create Account & Cluster:** Register at <https://www.mongodb.com/cloud/atlas>. Click **Create a Deployment** and select **M0 Free Tier**.
2. **Create Database User:** Navigate to **Security** → **Database Access**. Click **Add New User**. Set username (e.g. `evobin_user`) and password with **Read and write to any database** role.
3. **Configure Network Access:** Navigate to **Security** → **Network Access**. Click **Add IP Address** and select **Allow Access from Anywhere (0.0.0.0/0)**.
4. **Copy Connection URI:** Navigate to **Database** → **Connect** → **Drivers** and copy your URI:

mongodb+srv://evobin_user:<password>@cluster0.xxxxx.mongodb.net/evobin_db?retryWrites=true&w=majority
5. **Seed Initial Database (Optional):** Update backend/.env with your URI and run:

cd backend && python -m app.seed_data

Step 2: Render Deployment (FastAPI Backend)

1. Push updated project repository to GitHub.
2. Go to <https://dashboard.render.com/>, click **New +** → **Web Service**, and connect your GitHub repo.
3. Configure Service Parameters:

Setting	Value
Root Directory	backend
Runtime	Python 3
Build Command	pip install -r requirements.txt

Start Command	uvicorn app.main:app --host 0.0.0.0 --port \$PORT
Environment Variables	MONGO_URI = mongodb+srv://... DB_NAME = evobin_db

4. Click **Create Web Service**. Once live, test docs at <https://<your-render-url>.onrender.com/docs>.

Step 3: Vercel Deployment (React Frontend)

1. Log into <https://vercel.com/dashboard> and click **Add New...** → **Project**.

2. Import your GitHub repository and select framework settings:

Setting	Value
Framework Preset	Vite
Root Directory	frontend
Build Command	npm run build
Output Directory	dist
Environment Variable	VITE_API_BASE_URL = https://<your-render-url>.onrender.com

3. Click **Deploy**. Vercel will build and assign your live domain (e.g. <https://evobin-frontend.vercel.app>).

Post-Deployment Verification

• **Backend Health Check:** Open <https://<your-render-app>.onrender.com/>. Output should return status 'EvoBin Production API running'.

• **Frontend Login & Flow:** Open <https://<your-vercel-app>.vercel.app/>. Log in with admin credentials **admin@ewaste.com** / **admin123** to verify end-to-end connectivity.